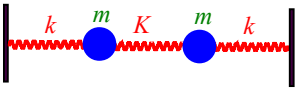
Homework 12, Week 15 2023.12.13 Due 2023.12.20 L05.03, L05.04, L05.06

★ L05.03

质量都是 m 的两个质点如图所知与三根水平放置的弹簧相连接，两侧弹簧的劲度系数为 k，中间弹簧劲度系数 K 试求解这两个质点水平方向的运动。



Solution

设两重点偏离平衡位形 $q^{(e)}$ 的小坐标分别为 r_1 和 r_2

体系的拉格朗日量：$L(r,\dot r) = T - U = \frac{1}{2} m \dot r_1^2 + \frac{1}{2} m \dot r_2^2 - \frac{1}{2} k r_1^2 - \frac{1}{2} k r_2^2 - \frac{1}{2} K (r_1 - r_2)^2$

$$\mathbb{T}_{ij} = \left.\frac{\partial^2 L(r,\dot r)}{\partial \dot r_i \partial \dot r_j}\right|_{r=0} \implies \mathbb{T} = \begin{pmatrix} m & 0 \\ 0 & m \end{pmatrix}, \qquad \mathbb{V}_{ij} = -\left.\frac{\partial^2 L(r,\dot r)}{\partial r_i \partial r_j}\right|_{r=0} \implies \mathbb{V} = \begin{pmatrix} K+k & -K \\ -K & k+K \end{pmatrix}$$

微振动频率：$|\mathbb{V} - \theta\,\mathbb{T}| = 0 \implies \begin{vmatrix} K+k-m\theta & -K \\ -K & k+K-m\theta \end{vmatrix} = 0 \implies \theta_{1,2} = \omega_{1,2}^2 = \frac{k}{m}, \ \frac{k+2K}{m}$

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eq = Det[{{K + k - m θ, -K}, {-K, K + k - m θ}}];
Solve[eq == 0, θ]
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$$\left\{\left\{\theta \to \frac{k}{m}\right\}, \left\{\theta \to \frac{k+2K}{m}\right\}\right\}$$

本征矢量：$\begin{pmatrix} K & -K \\ -K & K \end{pmatrix}\begin{pmatrix} z_1^{(1)} \\ z_2^{(1)} \end{pmatrix} = 0 \implies z^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

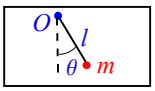
$$\begin{pmatrix} -K & -K \\ -K & -K \end{pmatrix}\begin{pmatrix} z_1^{(2)} \\ z_2^{(2)} \end{pmatrix} = 0 \implies z^{(2)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\mathbb{r} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}(A_1 \cos\omega_1 t + B_1 \sin\omega_1 t) + \begin{pmatrix} 1 \\ -1 \end{pmatrix}(A_2 \cos\omega_2 t + B_2 \sin\omega_2 t),$$

其中：$\omega_1 = \sqrt{\frac{k}{m}}$，$\omega_2 = \sqrt{\frac{k+2K}{m}}$，系数 A_1，B_1，A_2，B_2 由初始位移、速度确定。

★ L05.04

一滑块质量为 $2m$ 放在水平光滑平面上，滑块上 O 点系着一长度为 l 质量为 m 的单摆，假定为平面运动，求系统的小振动解



Solution

以小坐标 x，θ 为广义坐标，其中 x 为滑块偏离院士位置的位移

拉格朗日量：$L = \frac{1}{2}(2m)\dot x^2 + \frac{m}{2}\left(l\dot\theta\cos\theta + \dot x\right)^2 + \frac{m}{2}\left(l\dot\theta\sin\theta\right)^2 - l(1-\cos\theta)\,m\,g$

保留到 x，θ，$\dot x$，$\dot\theta$ 的二次项，$L = m\dot x^2 + \frac{m}{2}\left(l\dot\theta + \dot x\right)^2 + \frac{m}{2} l^2 \dot\theta^2 - \frac{1}{2} l \theta^2 m g$

$$\mathbb{T}_{ij} = \left.\frac{\partial^2 L(r,\dot r)}{\partial \dot r_i \partial \dot r_j}\right|_{r=0} \implies \mathbb{T} = \begin{pmatrix} 3m & ml \\ ml & ml^2 \end{pmatrix}, \qquad \mathbb{V}_{ij} = -\left.\frac{\partial^2 L(r,\dot r)}{\partial r_i \partial r_j}\right|_{r=0} \implies \mathbb{V} = \begin{pmatrix} 0 & 0 \\ 0 & mgl \end{pmatrix}$$

微振动频率：$|\mathbb{V} - \theta\,\mathbb{T}| = 0 \implies \begin{vmatrix} -3\,m\,\theta & -m\,l\,\theta \\ -m\,l\,\theta & m\,g\,l - m\,l^2\,\theta \end{vmatrix} = 0 \implies \theta_{1,2} = \omega_{1,2}^2 = 0,\ \dfrac{3\,g}{2\,l}$

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t = {{-3 m θ, -m l θ}, {-m l θ, m g l - m l l θ}};
eq = Det[t];
Solve[eq == 0, θ]
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$$\left\{\{\theta \to 0\},\ \left\{\theta \to \frac{3\,g}{2\,l}\right\}\right\}$$

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t /. θ → 3 g / 2 / l
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$$\left\{\left\{-\frac{9\,g\,m}{2\,l},\ -\frac{3\,g\,m}{2}\right\},\ \left\{-\frac{3\,g\,m}{2},\ -\frac{1}{2}\,g\,l\,m\right\}\right\}$$

本征矢量：$\begin{pmatrix} 0 & 0 \\ 0 & m\,g\,l \end{pmatrix}\begin{pmatrix} z_1^{(1)} \\ z_2^{(1)} \end{pmatrix} = 0 \quad \implies z^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

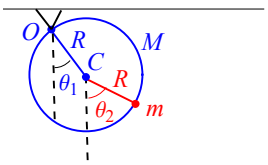
$$\frac{m\,g}{2}\begin{pmatrix} -9/l & -3 \\ -3 & -l \end{pmatrix}\begin{pmatrix} z_1^{(2)} \\ z_2^{(2)} \end{pmatrix} = 0 \quad \implies z^{(2)} = \begin{pmatrix} l/3 \\ -1 \end{pmatrix}$$

$$\mathbb{r} = \begin{pmatrix} x \\ \theta \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}(A_1 + B_1\,t) + \begin{pmatrix} l/3 \\ -1 \end{pmatrix}(A_2 \cos\omega_2\,t + B_2 \sin\omega_2\,t)$$

其中 $\omega_2 = \sqrt{\dfrac{3\,g}{2\,l}}$， 第一个简正模式是平动，质点 m 保持 $\theta = 0$

★ L05.06

匀质环质量为 M 半径为 R 可绕通过环上一点垂直于环面的水平轴自由转动。质量为 m 的质点穿在环上，可以自由滑动。写出系统的拉格朗日方程，求小振东解。



Solution

取 θ_1，θ_2 为广义小坐标，设拉格朗日量为 $L = T - V$

$$T = \frac{1}{2}\,I\,\dot\theta_1^2 + \frac{m}{2}\left(R\,\dot\theta_2 \cos\theta_2 + R\,\dot\theta_1 \cos\theta_1\right)^2 + \frac{m}{2}\left(R\,\dot\theta_2 \sin\theta_2 + R\,\dot\theta_1 \sin\theta_1\right)^2$$

$$V = M\,g\,R\,(1 - \cos\theta_1) + m\,g\,R\,[(1 - \cos\theta_1) + (1 - \cos\theta_2)]$$

其中 $I = M\,R^2 + M\,R^2 = 2\,M R^2$

保留到 θ_1，θ_2 的二次项

$$L = M R^2\,\dot\theta_1^2 + \frac{m}{2}\,R^2\left(\dot\theta_1^2 + \dot\theta_2^2 + 2\,\dot\theta_1\,\dot\theta_2\right) - \frac{M\,g\,R}{2}\,\theta_1^2 - \frac{m\,g\,R}{2}\left(\theta_1^2 + \theta_2^2\right)$$

$$\mathbb{T}_{i\,j} = \left.\frac{\partial^2 L(r,\dot r)}{\partial \dot r_i\,\partial \dot r_j}\right|_{r=0} \implies \mathbb{T} = R^2\begin{pmatrix} 2\,M + m & m \\ m & m \end{pmatrix}, \qquad \mathbb{V}_{i\,j} = -\left.\frac{\partial^2 L(r,\dot r)}{\partial r_i\,\partial r_j}\right|_{r=0} \implies \mathbb{V} = g\,R\begin{pmatrix} M + m & 0 \\ 0 & m \end{pmatrix}$$

微振动频率：$|\mathbb{V} - \theta\,\mathbb{T}| = 0 \implies \begin{vmatrix} (M + m)\,R\,g - \theta\,(2\,M + m)\,R^2 & -m\,R^2\,\theta \\ -m\,R^2\,\theta & m\,R\,g - m\,R^2\,\theta \end{vmatrix} = 0 \implies \theta_{1,2} = \omega_{1,2}^2 = \dfrac{g}{2\,R},\ \dfrac{(m + M)\,g}{M\,R}$

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t = {{(M + m) R g - (2 M + m) R^2 θ, -m R^2 θ}, {-m R^2 θ, m g R - m R^2 θ}};
eq = Det[t];
sol = Solve[eq == 0, θ]
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$$\left\{\left\{\theta \to \frac{g}{2R}\right\}, \left\{\theta \to \frac{g\,(m+M)}{M\,R}\right\}\right\}$$

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Simplify[t /. sol[[2]]]
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$$\left\{\left\{-\frac{g\,(m+M)^2\,R}{M}, -\frac{g\,m\,(m+M)\,R}{M}\right\}, \left\{-\frac{g\,m\,(m+M)\,R}{M}, -\frac{g\,m^2\,R}{M}\right\}\right\}$$

本征矢量：$\begin{pmatrix} m\,g\,R/2 & -m\,g\,R/2 \\ -m\,g\,R/2 & m\,g\,R/2 \end{pmatrix}\begin{pmatrix} z_1^{(1)} \\ z_2^{(1)} \end{pmatrix} = 0 \quad \Longrightarrow \quad z^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$-\frac{R\,g}{M}\begin{pmatrix} (m+M)^2 & m(m+M) \\ m(m+M) & m^2 \end{pmatrix}\begin{pmatrix} z_1^{(2)} \\ z_2^{(2)} \end{pmatrix} = 0 \quad \Longrightarrow \quad z^{(2)} = \begin{pmatrix} m/(M+m) \\ -1 \end{pmatrix}$$

$$\mathbb{r} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}(A_1 \cos\omega_1 t + B_1 \sin\omega_1 t) + \begin{pmatrix} m/(M+m) \\ -1 \end{pmatrix}(A_2 \cos\omega_2 t + B_2 \sin\omega_2 t)$$

其中：$\omega_1 = \sqrt{\dfrac{g}{2R}}, \quad \omega_2 = \sqrt{\dfrac{(m+M)\,g}{M\,R}}$